



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Piotr Kiciński
July, 2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Methodology: collected data of launches with SpaceX REST API and additionally by Wikipedia web scraping, wrangled the labeled 90 Falcon 9 launches, then applied EDA (including visualization and some SQL queries), then performed interactive geospatial analysis with the use of Folium library and an interactive dashboard built-up using Plotly Dash library, and then supervised a few classification models to predict first-stage landing success.
- Results: launch successful attempts has gone up from near 0% in 2013 to over 80% in the recent period; Out of all launch sites the KSC LC-39A shows the highest site-level success rate; out of available and analyzed features payload mass and orbit type are strongest predictors of success in landing; the best classification model (Decision Tree) reached 88.9% cross-validation accuracy and 83.3% test accuracy.

Introduction

- SpaceX advertises Falcon 9 launches at \$62M cost, versus \$165M+ charged by competitors— largely because reuse of the rocket's first stage instead of discarding it.
- If we can predict whether the first stage will land successfully, we can estimate the true cost of a launch — valuable for any competitor bidding against SpaceX, and for SpaceX itself in planning reuse.
- This project aims at finding solutions to a problem: which factors most strongly predict a successful first-stage landing, and can a classification model predict this outcome reliably from historical data?

Section 1

Methodology

Methodology

Data collection methodology:

- Collections of 90 Falcon 9 launch records via the SpaceX REST API and by web scraping the Wikipedia "List of Falcon 9 and Falcon Heavy launches" page with BeautifulSoup
- Performed data wrangling: handled missing values, engineered the binary landing-success "Class" label from mission outcome text
- Performed exploratory data analysis (EDA) using visualization (Seaborn/Matplotlib) and SQL (SQLite) to surface patterns by site, orbit, payload and year
- Performed interactive visual analytics using Folium (launch-site geospatial maps) and Plotly Dash (interactive success-rate dashboard)
- Performed predictive analysis using classification models (Logistic Regression, SVM, Decision Tree, KNN) — built, hyperparameter-tuned via GridSearchCV, evaluated, and compared to find the best performer

Data Collection

- Two complementary data sources were combined into a single launch dataset:
- **1) SpaceX REST API (api.spacexdata.com)**
- GET requests against the /launches, /rockets, /launchpads and /payloads endpoints; JSON responses parsed and joined into a Pandas DataFrame covering flight number, date, booster version, payload mass, orbit, launch site and outcome
- **2) Wikipedia web scraping**
- Historical Falcon 9 / Falcon Heavy launch table scraped from the Wikipedia page "List of Falcon 9 and Falcon Heavy launches" using BeautifulSoup, parsed into a matching Pandas DataFrame as a cross-check / supplementary source
- Combined result: 90 Falcon 9 launch records used for downstream wrangling and modelling

Data Collection – SpaceX API

- Sent GET requests to the SpaceX REST API (api.spacexdata.com) to retrieve launch, rocket, launchpad and payload records as JSON
- Normalized nested JSON into flat columns (flight number, date, booster version, payload mass, orbit, launch site) and merged into a single Pandas DataFrame
- GitHub URL:

[https://github.com/piokici/IBM-DataScience-SpaceX-Capstone/blob/main/jupyter-labs-spacex-data-collection-api%20\(1\).ipynb](https://github.com/piokici/IBM-DataScience-SpaceX-Capstone/blob/main/jupyter-labs-spacex-data-collection-api%20(1).ipynb)

- Flowchart Schema:
- A[Start: Import past launches] --> B[GET https://api.spacexdata.com/v4/launches/past]
- B --> C[JSON conversions to → DataFrame (pd.json_normalize)]
- C --> D[Data Filtering] D --> D1[Reject flights with z >1 core] D --> D2[Reject flights with z >1 payload] D --> D3[Keep Falcon 9 Only] D --> D4[Dates ≤ 2020-11-13]
- D --> E[Preparation of auxiliary list]
- E --> F1[getBoosterVersion GET /rockets/{rocket_id}] E --> F2[getLaunchSite GET /launchpads/{launchpad_id}] E --> F3[getPayloadData GET /payloads/{payload_id}] E --> F4[getCoreData GET /cores/{core_id}]
- F1 & F2 & F3 & F4 --> G[Making launch_dict]
- G --> H[Final DataFrame]
- H --> I[Export to CSV: dataset_part_1.csv]
- I --> J[End]

Data Collection - Scraping

- Requested the static Wikipedia snapshot (List of Falcon 9 and Falcon Heavy launches, 9 June 2021 revision) via `requests.get()` with a browser User-Agent header
- Parsed the HTML with BeautifulSoup; located the launch-records table (3rd `<table>` on the page) and extracted column headers from its `<th>` elements
- Iterated table rows to fill a `launch_dict` (flight no., launch site, payload, payload mass, orbit, customer, outcome), then converted to a Pandas DataFrame
- GitHub:

<https://github.com/piokici/IBM-DataScience-SpaceX-Capstone/blob/main/jupyter-labs-webscraping.ipynb>

Data Wrangling

- Loaded the 90-row combined launch dataset (dataset_part_1.csv) and profiled missing values
- Identified numerical vs. categorical dtypes for all columns to prepare for downstream encoding
- Counted launches per site: CCAFS SLC-40 (55), KSC LC-39A (22), VAFB SLC-4E (13)
- Counted occurrences of each orbit type (excluding GTO as a transfer orbit): ISS (21), VLEO (14), PO (9), LEO (7), SSO (5), MEO (3), plus single-count orbits
- Engineered the binary target label "Class" from the Outcome column: 0 for unsuccessful landings (False Ocean/RTLS/ASDS, None), 1 for successful — overall success rate = 66.7%
- Exported the labeled dataset to CSV (dataset_part_2.csv) for the EDA and modelling stages

EDA with Data Visualization

- Flight Number vs. Payload Mass (catplot, hue=Class): later flights land successfully more often, and heavier payloads don't prevent success
- Flight Number vs. Launch Site (catplot): success rate visibly improves over time at every site
- Payload Mass vs. Launch Site (scatter): VAFB SLC-4E never launched a payload above ~10,000 kg
- Success Rate vs. Orbit Type (bar chart): ES-L1, GEO, HEO, SSO and VLEO orbits show the highest success rates
- Flight Number vs. Orbit Type (scatter): LEO success correlates with flight number; GTO shows no such relationship
- Launch Success Yearly Trend (line chart): success rate rose steadily from 2013 through 2020

EDA with SQL

- Loaded the SpaceX dataset into a SQLite database (my_data1.db) and queried it with 10 SQL tasks covering:
- Unique launch site names, and records filtered by site-name prefix
- Aggregate payload mass by customer (NASA) and by booster version (F9 v1.1)
- Date of first successful ground-pad landing, and boosters landing on a drone ship within a payload range
- Total counts of successful vs. failed mission outcomes
- Booster(s) that carried the maximum payload mass
- Month-level breakdown of 2015 landing failures, and a ranked count of landing outcomes over a multi-year window

Build an Interactive Map with Folium

- `folium.Circle` + `folium.Marker` for each of the 4 launch sites (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E), labelled with `DivIcon` text, centred on an initial view of NASA Johnson Space Center
- `MarkerCluster` of all 90 individual launch records, colour-coded green (success) / red (failure) via a `marker_color` column derived from `Class` — lets you see each site's success mix at a glance
- `MousePosition` plugin added to read live lat/long while hovering, used to pick coastline/city reference points
- `folium.PolyLine` + distance calculation (haversine formula) from KSC LC-39A to its nearest coastline point (~0.9 km) — added as a proximity example
- Github: https://github.com/piokici/IBM-DataScience-SpaceX-Capstone/blob/main/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

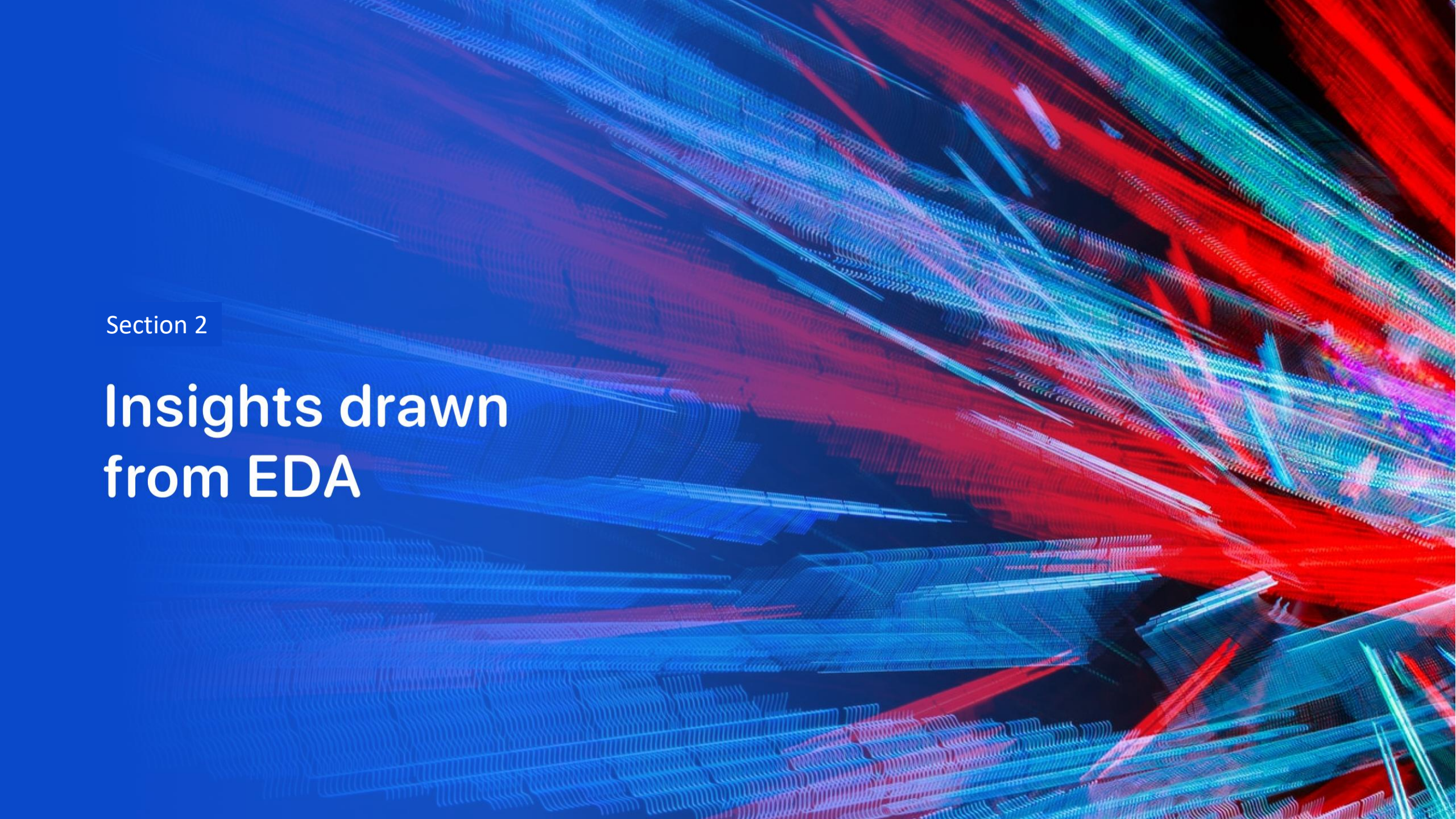
- Dashboard design (site-dropdown + payload-slider inputs) planned around two callbacks: a launch-success pie chart and a payload-vs-outcome scatter plot
- Pie chart: shows total successful launches by site when "All Sites" is selected, or the success/failure split for one selected site
- Scatter chart: Payload Mass vs. launch outcome (Class), coloured by Booster Version Category, filtered live by the payload-mass range slider and the selected site
- Interactions let a reviewer isolate, e.g., "which payload range has the best success rate at KSC LC-39A" in a couple of clicks

Predictive Analysis (Classification)

- Built the feature matrix X (80 columns after one-hot encoding Orbit, LaunchSite, LandingPad, Serial) and target Y (Class), standardized X with StandardScaler
- Split into train/test with train_test_split (test_size=0.2, random_state=2) — 72 training rows, 18 test rows
- Trained 4 classifiers, tuning hyperparameters via GridSearchCV (cv=10, scoring='accuracy') for each: Logistic Regression, SVM, Decision Tree, KNN
- Compared best cross-validation accuracy across models, then evaluated the tuned models on the held-out test set and inspected confusion matrices
- Selected the Decision Tree as the best performer (88.9% CV accuracy) and used its confusion matrix to characterize remaining errors

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

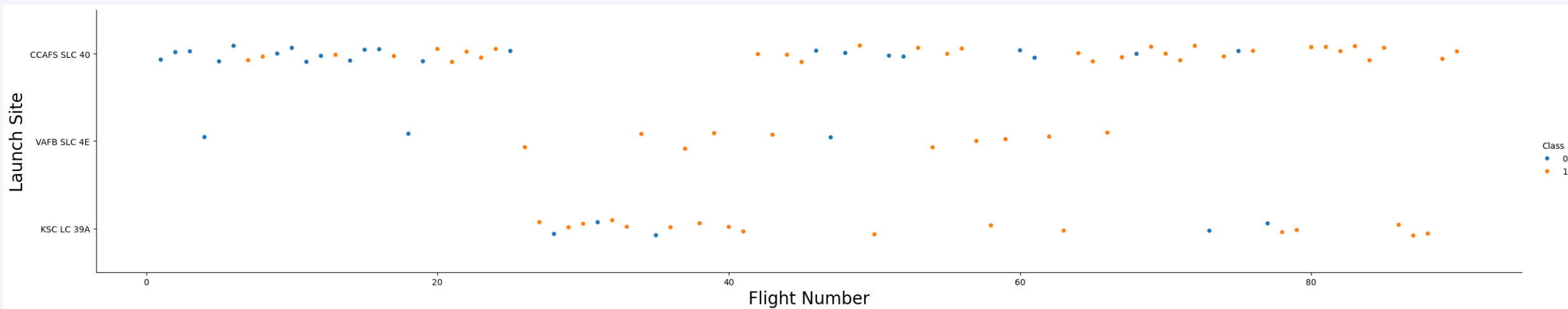
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

Insights drawn from EDA

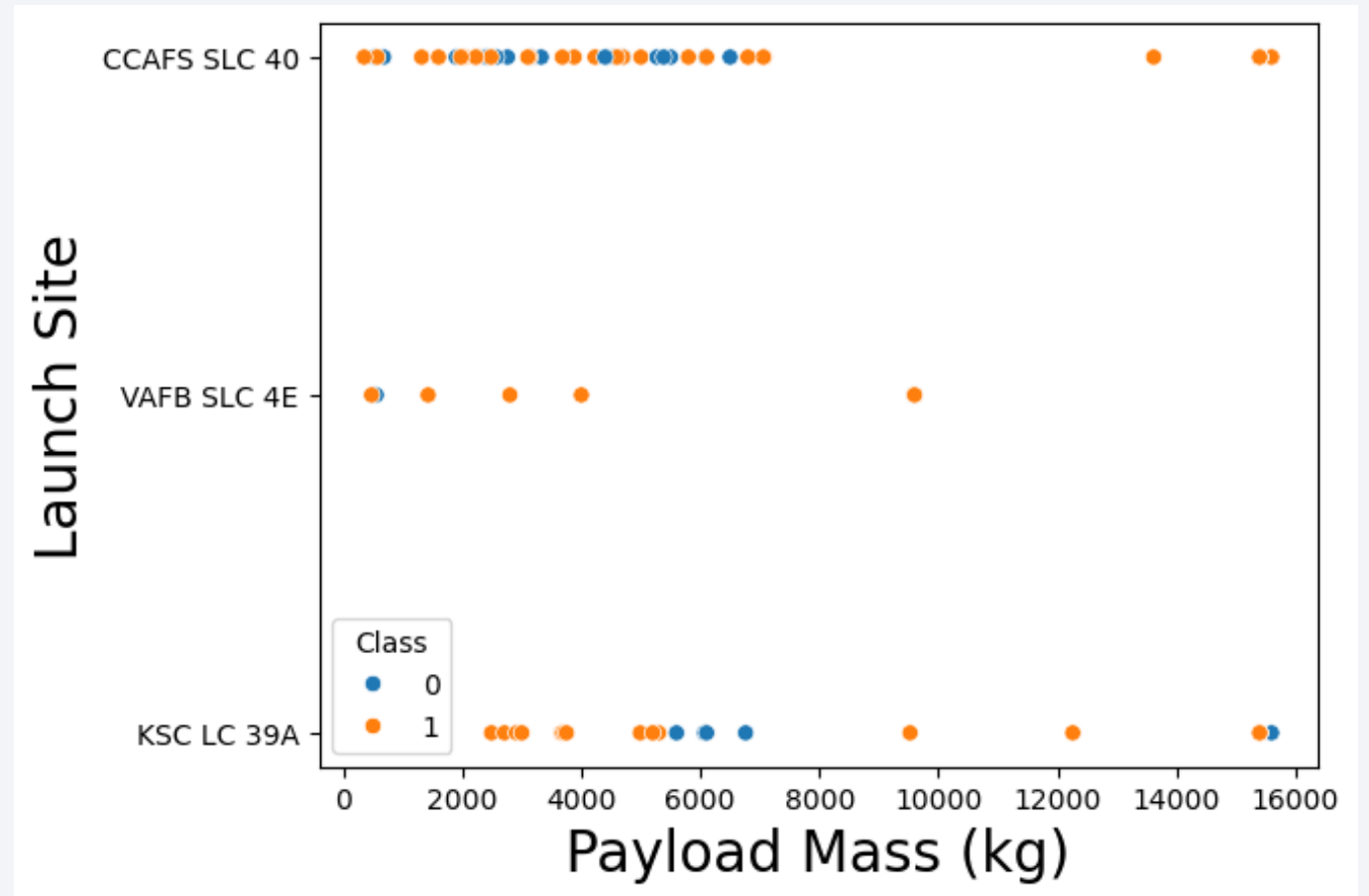
Flight Number vs. Launch Site

- As flight number increases over time, the proportion of successful landings (green) rises sharply at every launch site
- Early flights (low flight number) cluster toward failures; later flights are dominated by successes — strong evidence of a learning-curve effect
- KSC LC-39A and CCAFS SLC-40 show the clearest late-flight success clustering



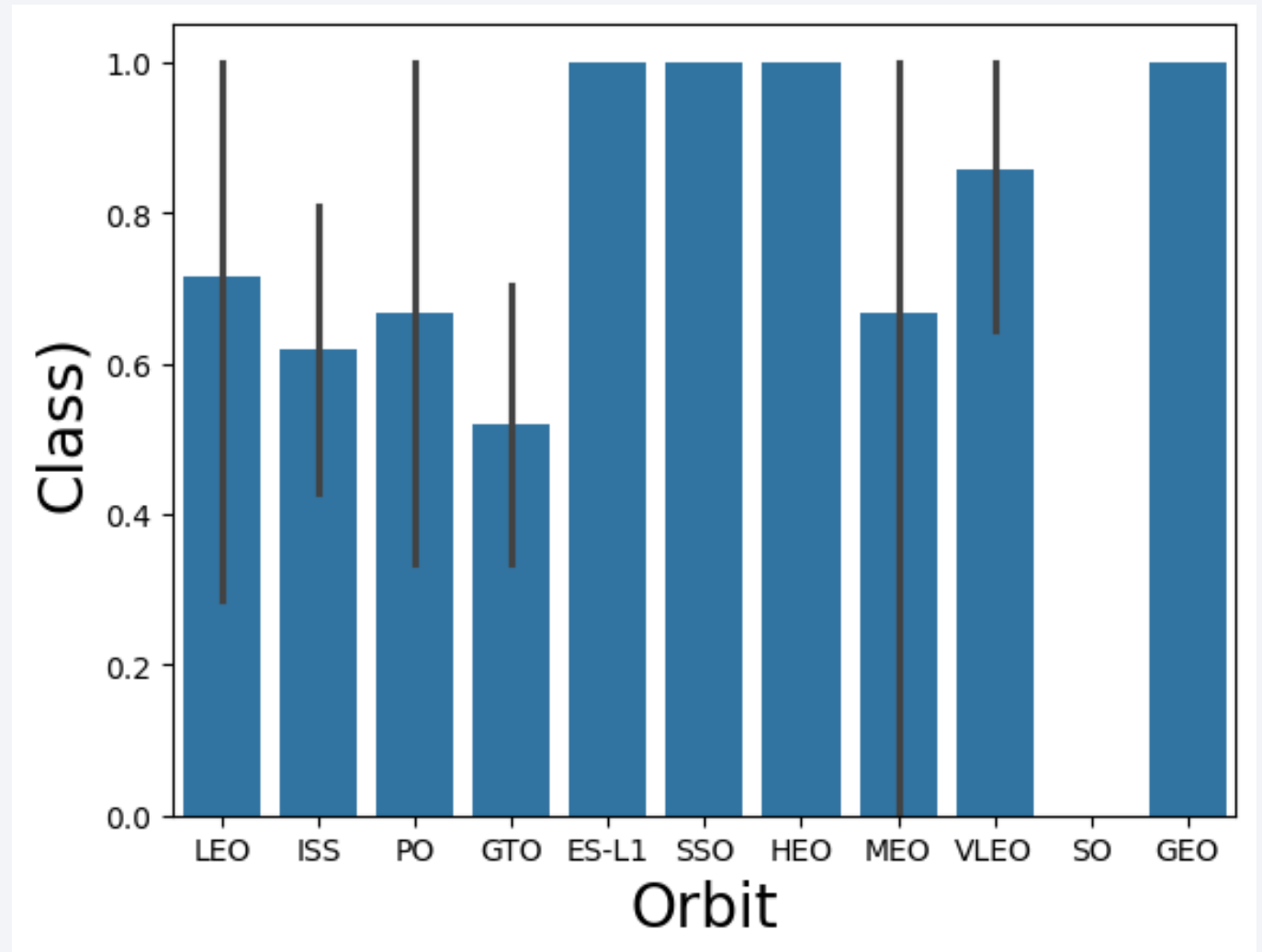
Payload vs. Launch Site

- VAFB SLC-4E never carried a payload heavier than ~10,000 kg — a clear site-level payload ceiling
- CCAFS SLC-40 and KSC LC-39A both handle the full payload range, including the heaviest missions
- No single site shows a hard payload cutoff for success — heavy payloads still land successfully at CCAFS/KSC



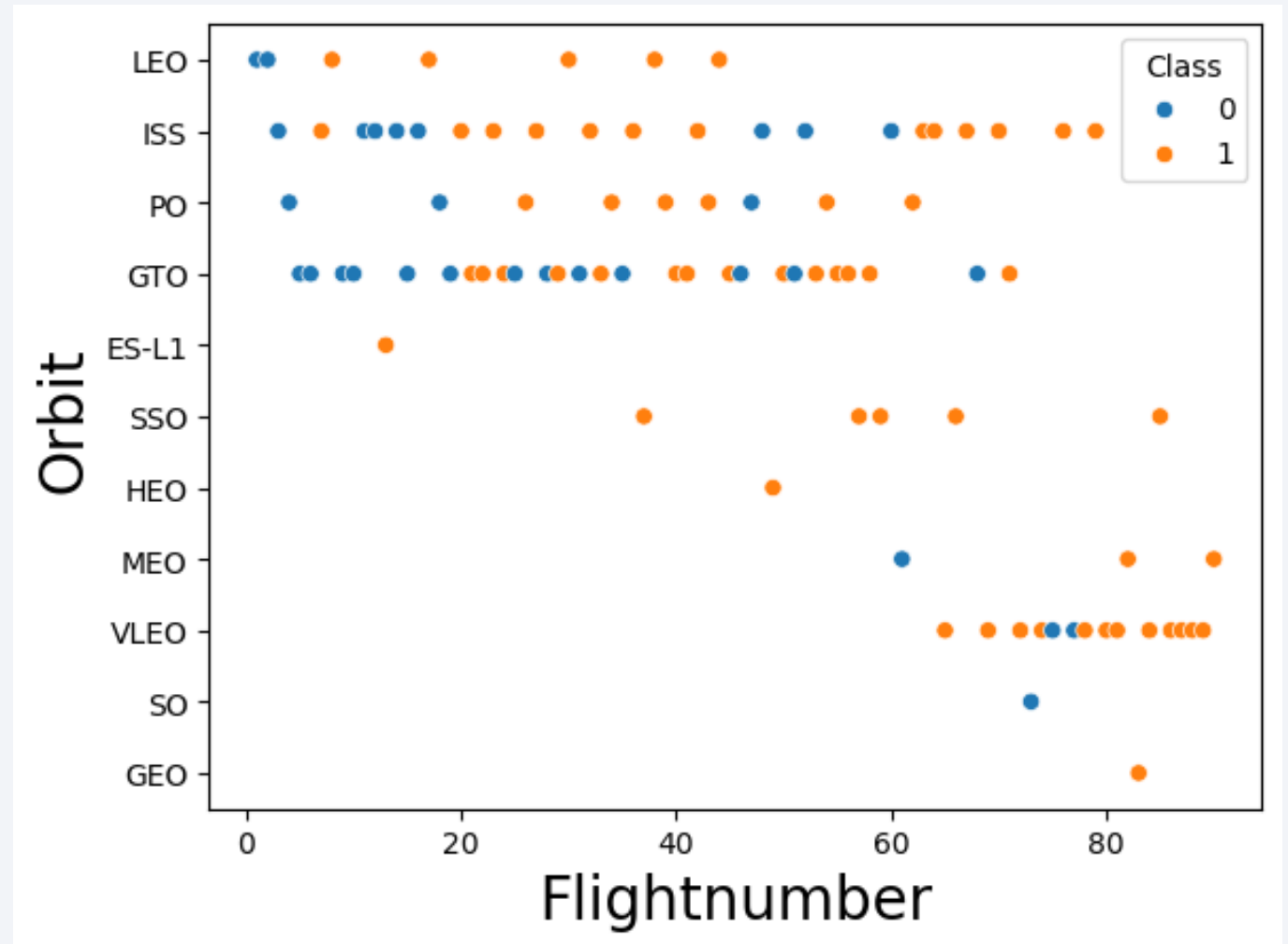
Success Rate vs. Orbit Type

- ES-L1, GEO, HEO, SSO and VLEO orbits show the highest success rates (at or near 100%)
- GTO and ISS orbits show more mixed outcomes — these tend to be higher-difficulty or higher-payload missions
- Orbit type is one of the strongest categorical predictors of landing success



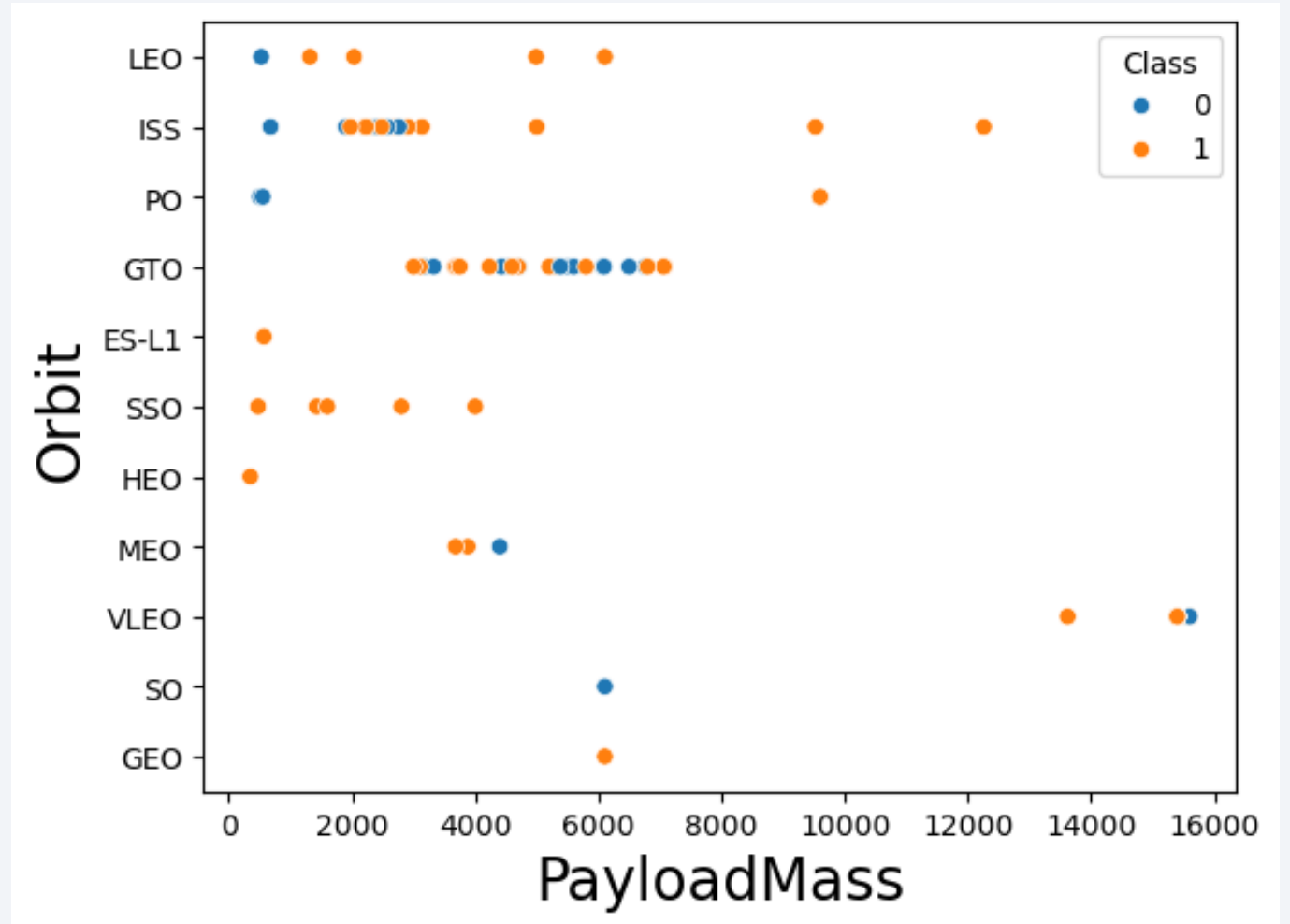
Flight Number vs. Orbit Type

- In LEO, success is clearly tied to flight number — later LEO flights succeed much more consistently
- In GTO, no such relationship appears — flight number alone doesn't explain GTO outcomes
- Suggests the model should treat orbit-specific learning curves differently, not a single global trend



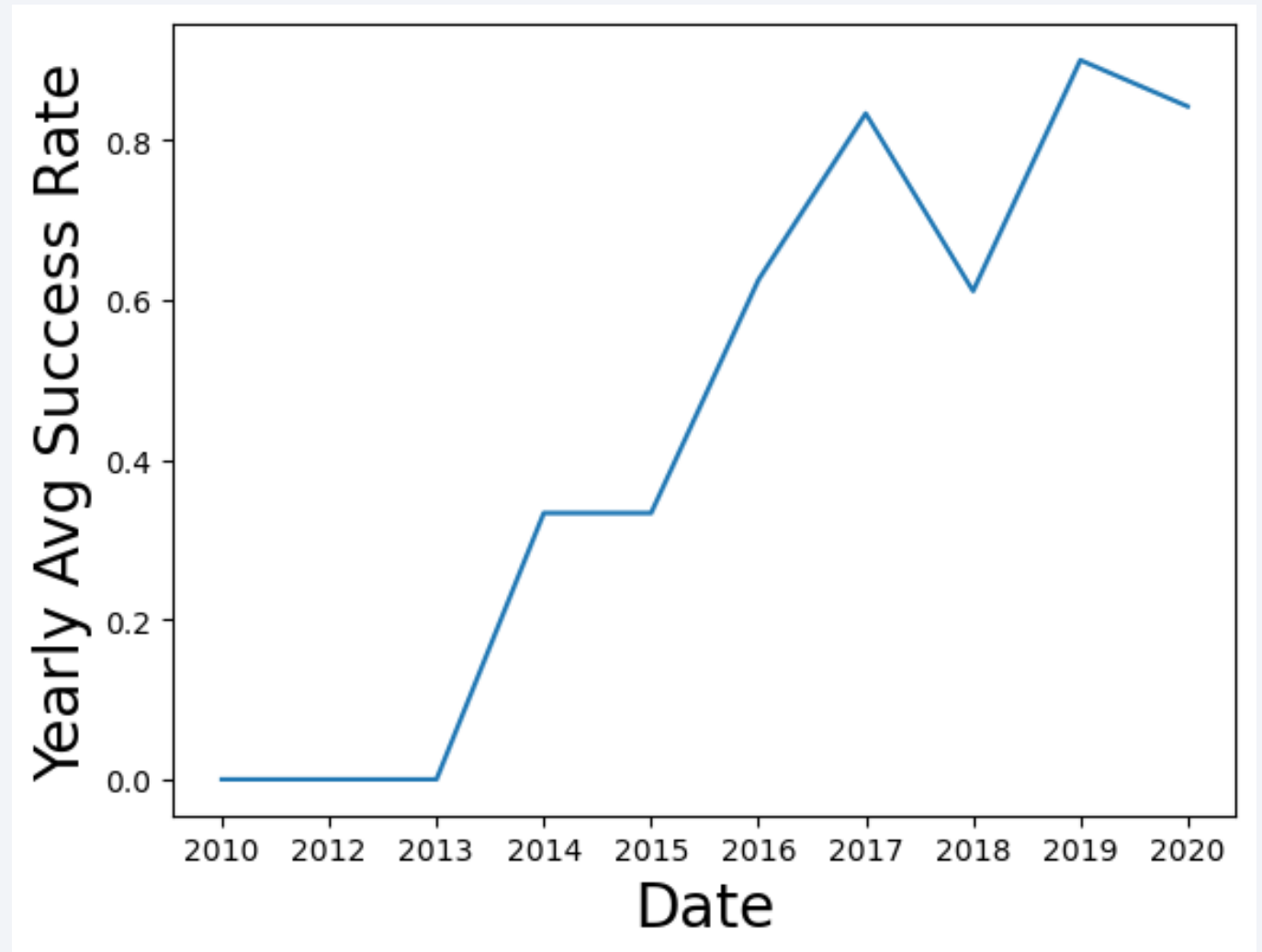
Payload vs. Orbit Type

- For Polar, LEO and ISS orbits, heavier payloads are associated with more successful landings
- For GTO, both successful and unsuccessful landings occur across the payload range — payload mass alone doesn't separate outcomes here
- Reinforces that Orbit and PayloadMass interact rather than acting independently



Launch Success Yearly Trend

- Yearly average success rate rose steadily from 2013 through 2020
- The steepest improvement occurs 2013-2017, consistent with SpaceX's public track record of iterating on booster recovery technique
- By 2019-2020 the yearly average success rate sits well above 80%



All Launch Site Names

- **SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE**
- Result: CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40
- Four distinct launch sites appear in the dataset — two at Cape Canaveral (LC-40, SLC-40), one at Kennedy Space Center (LC-39A), one at Vandenberg (SLC-4E)

Launch Site Names Begin with 'CCA'

- **SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5**
- Returned the first 5 Cape Canaveral launches, starting with the 2010-06-04 Falcon 9 v1.0 debut flight (Dragon Spacecraft Qualification Unit, LEO, Success/Failure(parachute))
- Confirms Cape Canaveral (CCAFS LC-40 / SLC-40) hosted SpaceX's earliest missions

Total Payload Mass

- `SELECT SUM("PAYLOAD_MASS__KG_") FROM SPACEXTABLE WHERE Customer='NASA (CRS)'`
- Result: 45,596 kg total payload mass carried for NASA under Commercial Resupply Services contracts

Average Payload Mass by F9 v1.1

- `SELECT AVG("PAYLOAD_MASS__KG_") FROM SPACEXTABLE WHERE "Booster_Version" LIKE 'F9 v1.1%'`
- Result: 2,534.67 kg average payload mass for the F9 v1.1 booster version

First Successful Ground Landing Date

- `SELECT MIN(Date), "Landing_Outcome" FROM SPACEXTABLE GROUP BY "Landing_Outcome"`
- Result: first ground-pad success ("Success (ground pad)") occurred on 2015-12-22
- For reference, the first ocean-landing success ("Controlled (ocean)") came earlier, on 2014-04-18

Successful Drone Ship Landing with Payload between 4000 and 6000

- **SELECT "Booster_Version" FROM SPACEXTABLE WHERE "Landing_Outcome"='Success (drone ship)' AND "PAYLOAD_MASS__KG_" BETWEEN 4000 AND 6000**
- Result: F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2
- Four boosters successfully landed on a drone ship carrying a mid-range (4,000-6,000 kg) payload

Total Number of Successful and Failure Mission Outcomes

- `SELECT COUNT(*), "Mission_Outcome" FROM SPACEXTABLE GROUP BY "Mission_Outcome"`
- Result: 98 "Success", 1 "Success (payload status unclear)", 1 "Success " (trailing-space variant), 1 "Failure (in flight)"
- Mission-level (payload delivery) success rate is over 99% — markedly higher than the first-stage landing success rate, since a booster landing failure doesn't necessarily doom the payload mission

Boosters Carried Maximum Payload

- `SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS__KG_" IN (SELECT MAX("PAYLOAD_MASS__KG_") FROM SPACEXTABLE)`
- Result: 12 Block 5 boosters tied for the maximum payload mass, incl. B1048, B1049, B1051, B1056, B1058 and B1060 (various flight numbers)
- All top-payload missions were flown on the newest Block 5 booster generation

2015 Launch Records

- `SELECT SUBSTR(Date,6,2) AS MONTH, "Landing_Outcome", "Booster_Version", "Launch_Site" FROM SPACEXTABLE WHERE DATE BETWEEN '2015-01-01' AND '2015-12-31'`
- Result: 2 drone-ship landing failures in 2015 — January (F9 v1.1 B1012) and April (F9 v1.1 B1015), both launched from CCAFS LC-40
- 2015 was still an early, high-failure-rate period for SpaceX's drone-ship landing attempts

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- **SELECT "Landing_Outcome", COUNT(*) FROM SPACEXTABLE WHERE DATE BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY "Landing_Outcome" ORDER BY COUNT(*) DESC**
- Result (top): No attempt (10), Success (drone ship) (5), Failure (drone ship) (5), Success (ground pad) (3), Controlled (ocean) (3)
- In this early window, "no landing attempt" was still the single most common outcome — landing attempts only became the default later in the program

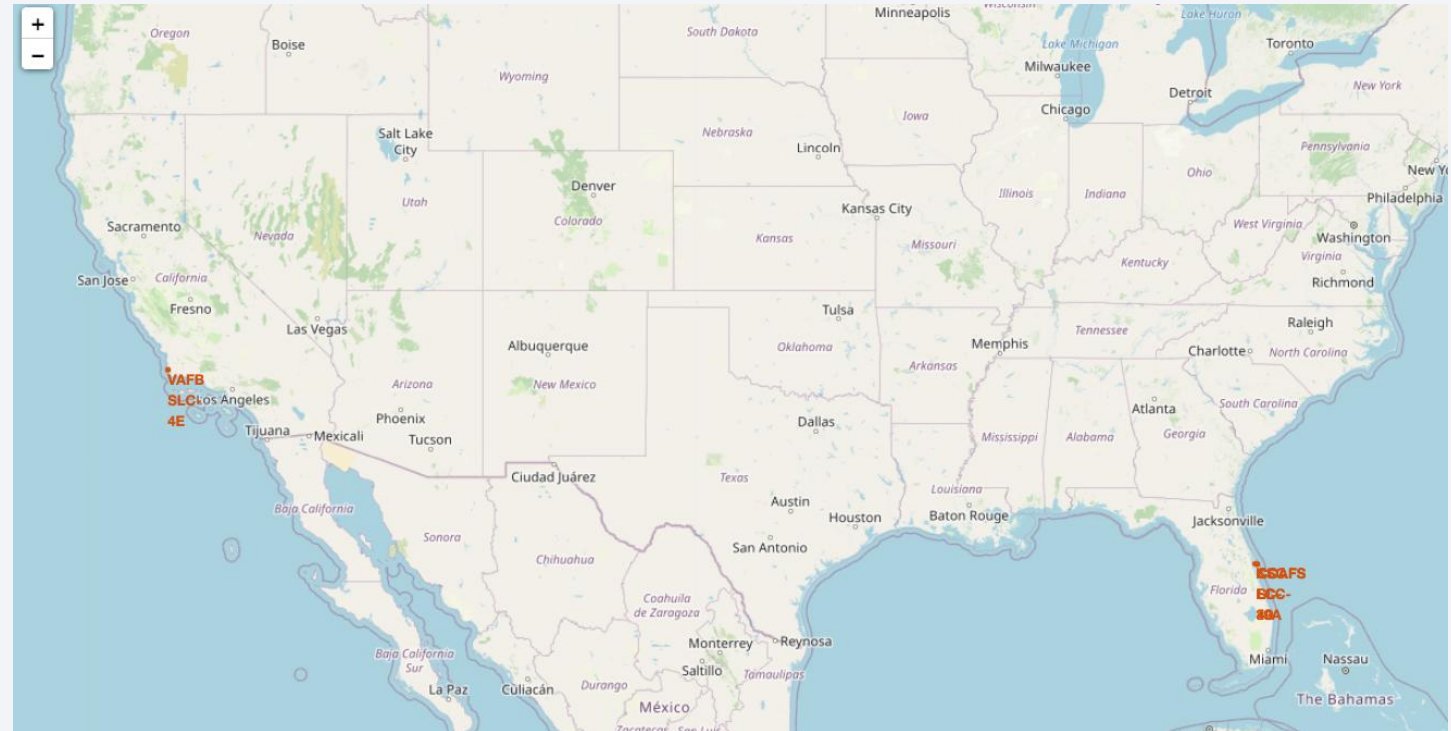
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

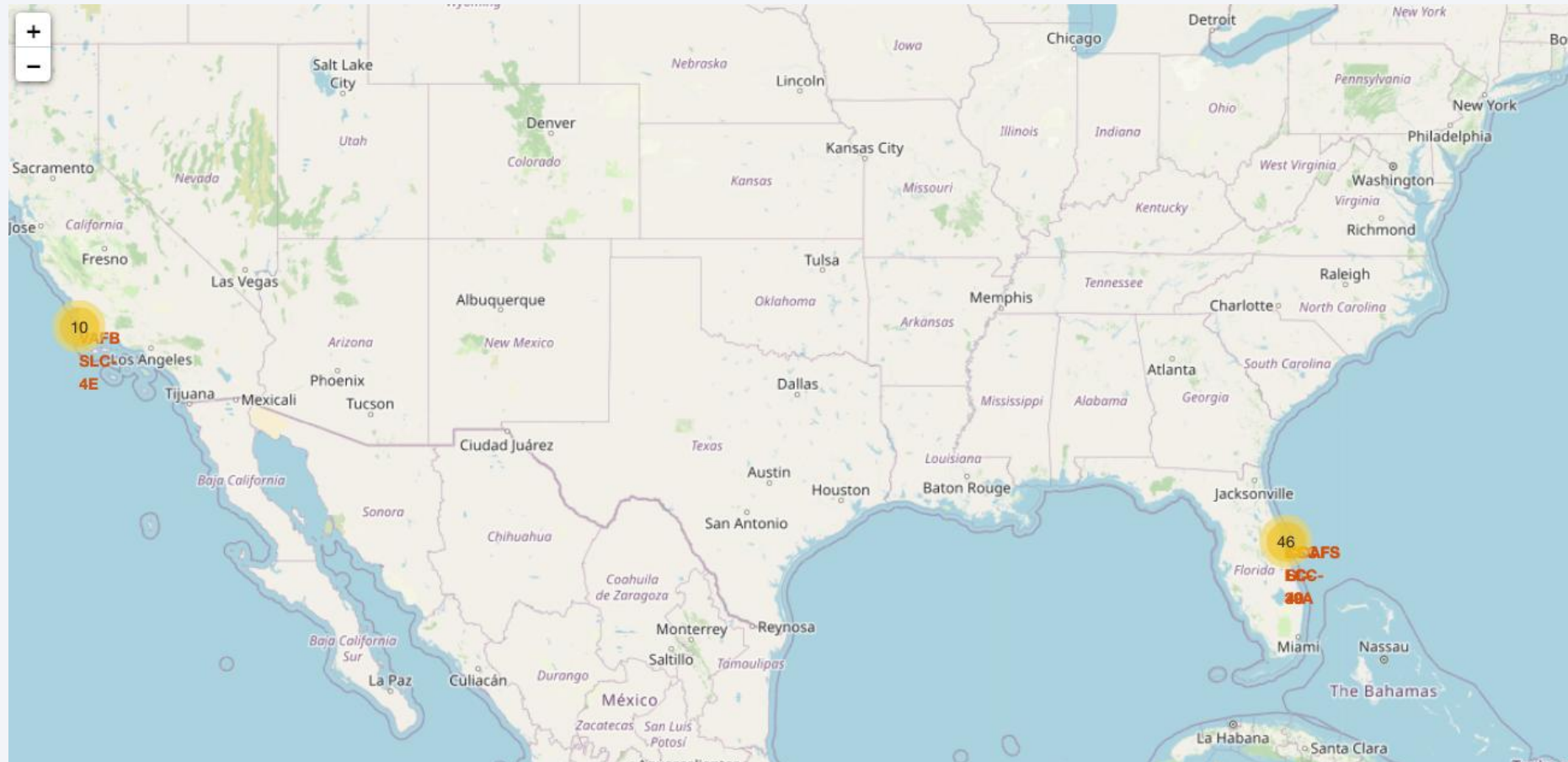
All Launch Sites on a Global Map

- [Map built with folium.Circle + folium.Marker at each site's (Lat, Long): CCAFS LC-40 (28.5623, -80.5774), CCAFS SLC-40 (28.5632, -80.5768), KSC LC-39A (28.5733, -80.6469), VAFB SLC-4E (34.6328, -120.6107)
- All Florida sites cluster tightly on the Atlantic coast near the equator-facing Cape Canaveral; VAFB SLC-4E sits alone on the California Pacific coast for polar/SSO launches



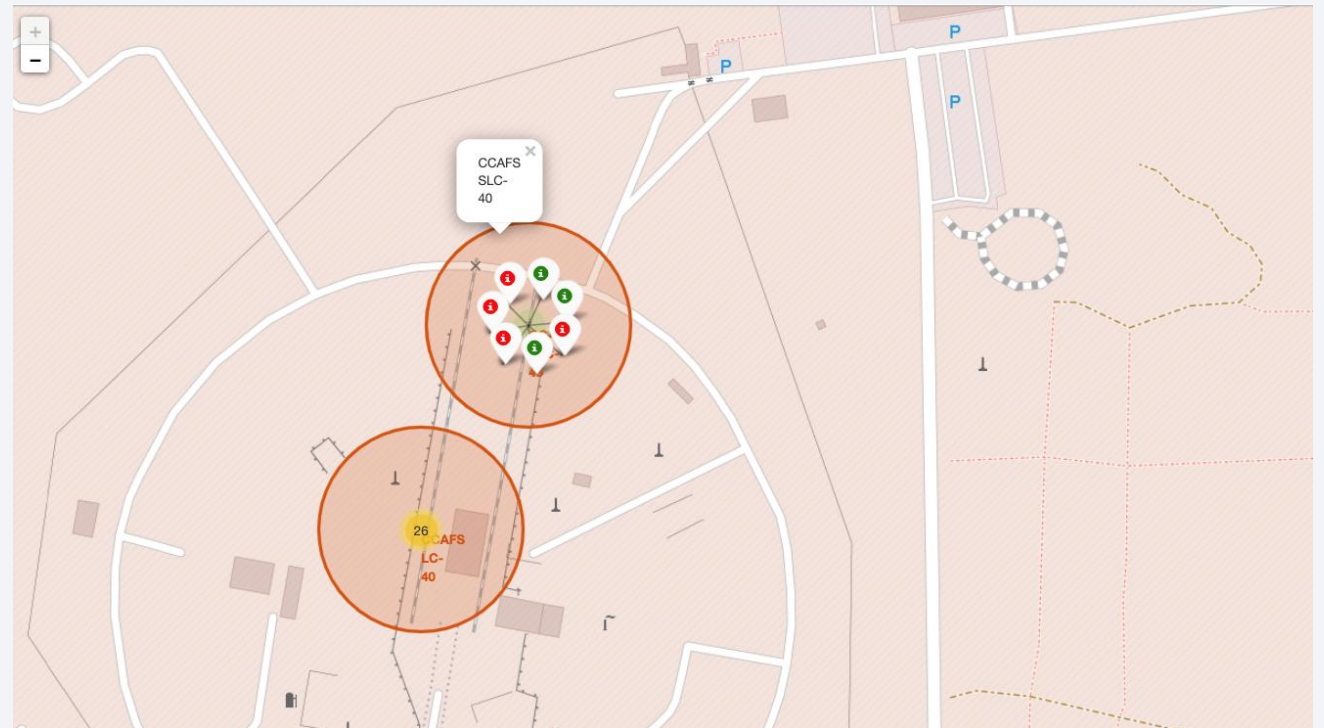
Color-Labeled Launch Outcomes by Site

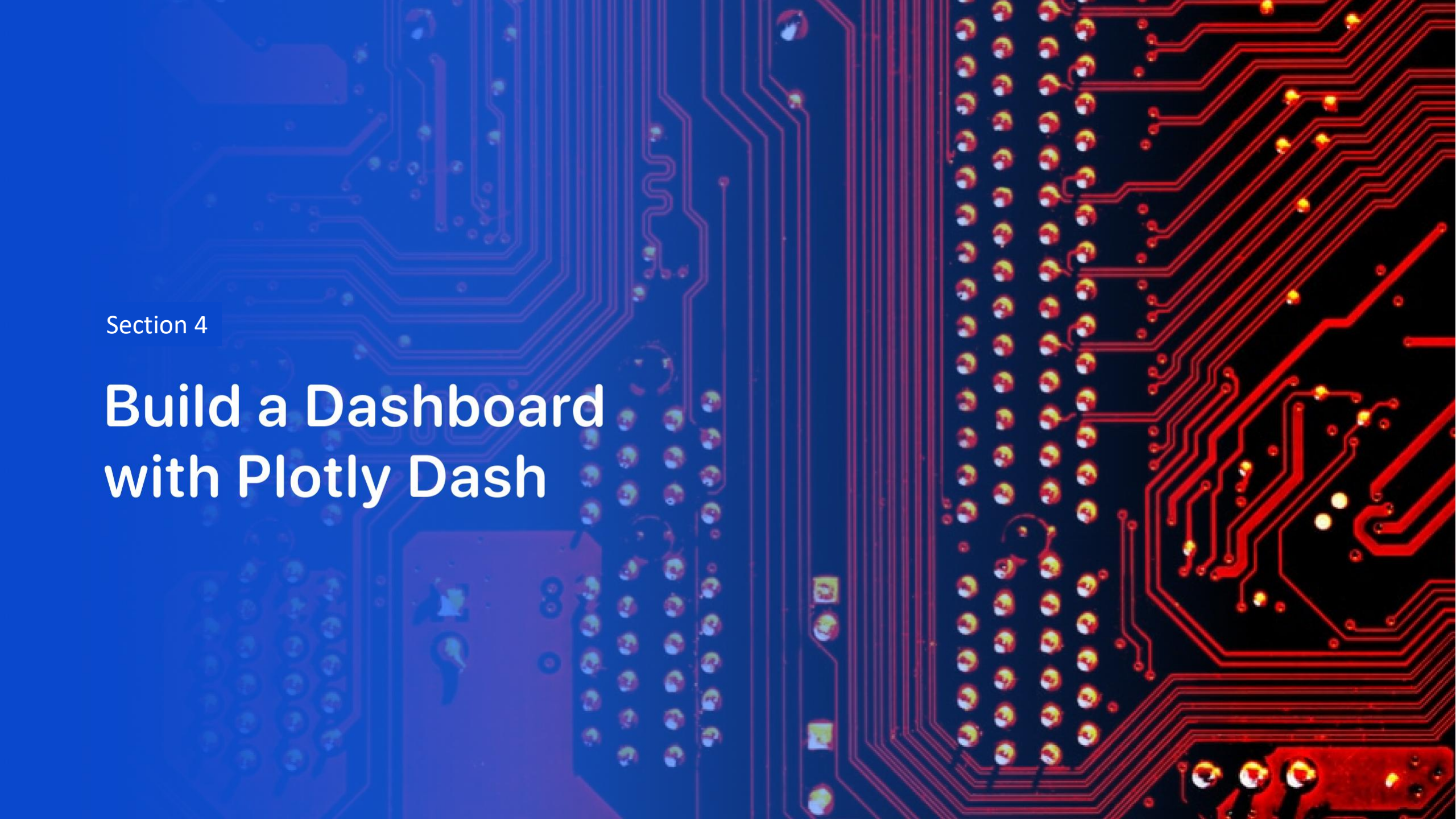
- [Each of the 90 launch records plotted as a green (success, Class=1) or red (failure, Class=0) marker inside a MarkerCluster per site
- Visually confirms the EDA finding: newer, high-volume sites (KSC LC-39A, CCAFS SLC-40) show cluster markers that are overwhelmingly green



KSC LC-39A — Proximity to the Coastline

- plugin used to read coordinates of the nearest coastline point; folium.PolyLine drawn from the launch site to that point
- Haversine distance calculated in the notebook between the launch-site coordinate and the coastline point along the Cape Canaveral shoreline
- Launch sites are built immediately adjacent to open water — consistent with range-safety requirements for a failed launch to fall over the ocean, not populated land



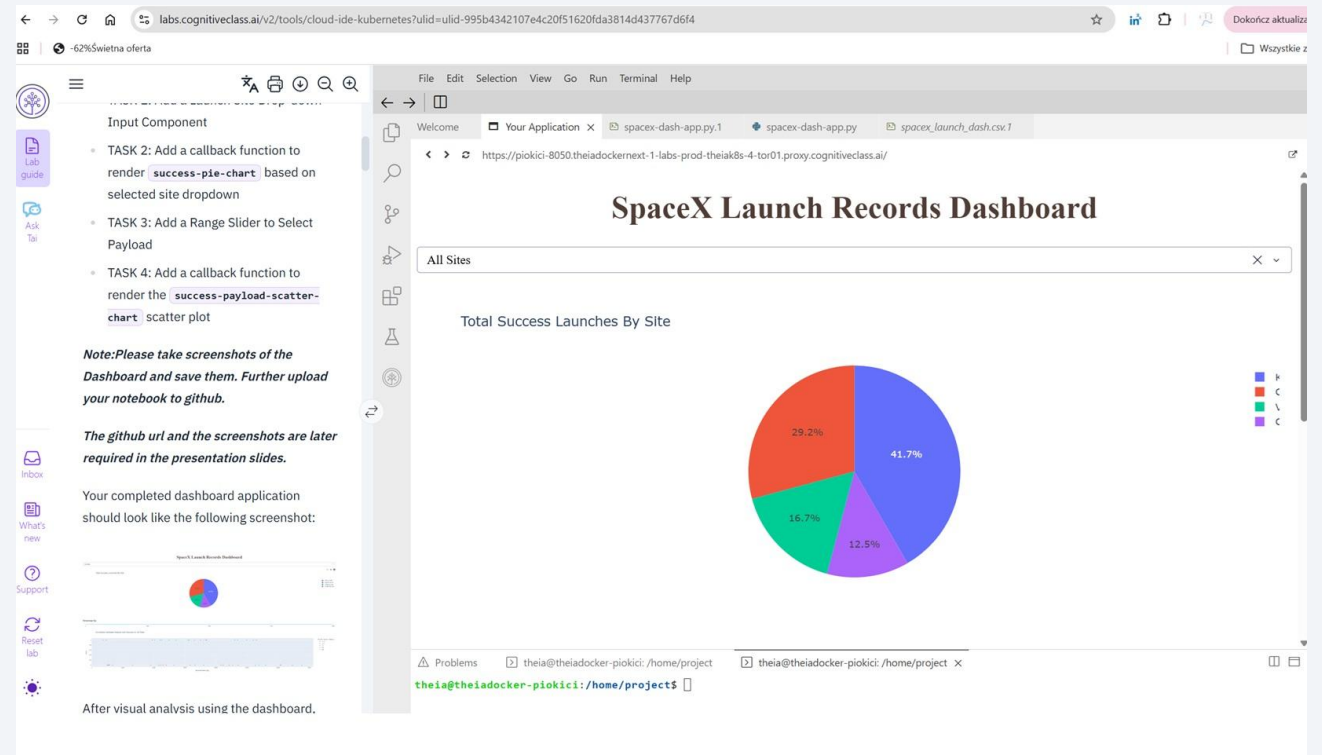


Section 4

Build a Dashboard with Plotly Dash

Launch Success Count — All Sites

- Pie chart built from `spacex_df` filtered to `Class=1`, grouped by Launch Site, via the site-dropdown -> `success-pie-chart` callback
- Expected pattern from the underlying data: CCAFS SLC-40 and KSC LC-39A contribute the largest shares of total successful launches, reflecting their higher launch volumes

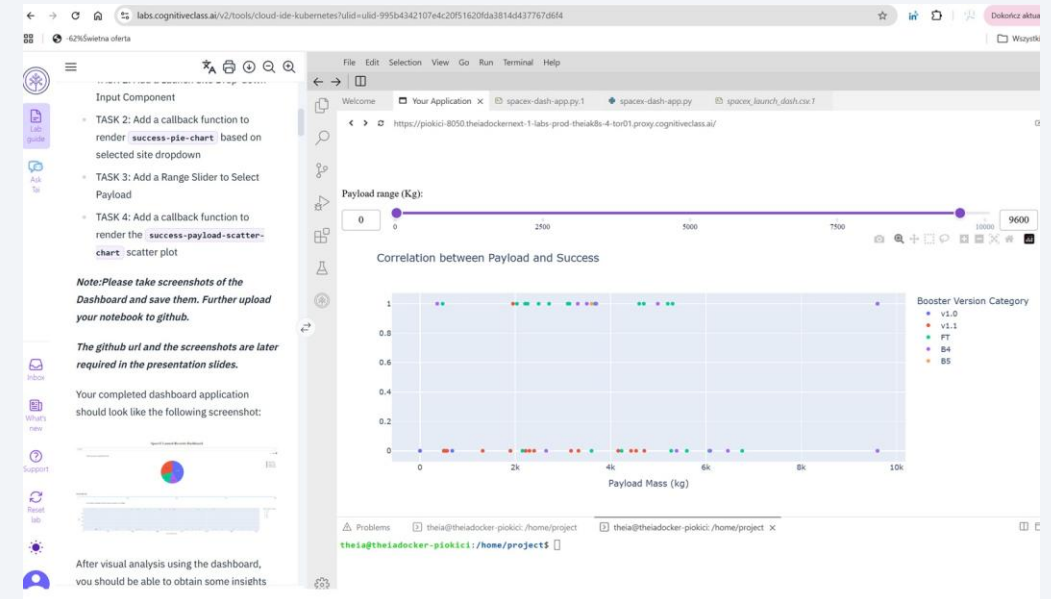


Highest Success-Ratio Site — KSC LC-39A

- When site-dropdown is set to a single site, the callback re-renders the pie chart as that site's own Success vs. Failure split
- Based on the EDA results, KSC LC-39A is expected to show the strongest success ratio among the four sites

Payload vs. Launch Outcome — Filtered by Range Slider

- success-payload-scatter-chart callback filters spacex_df by the payload-slider range and by the selected site, plotting Payload Mass vs. Class coloured by Booster Version Category
- Expected finding, consistent with the static EDA: the 2,000-4,000 kg payload band shows the highest success density; Booster Version FT and Block 5 categories cluster toward more recent, higher-success launches

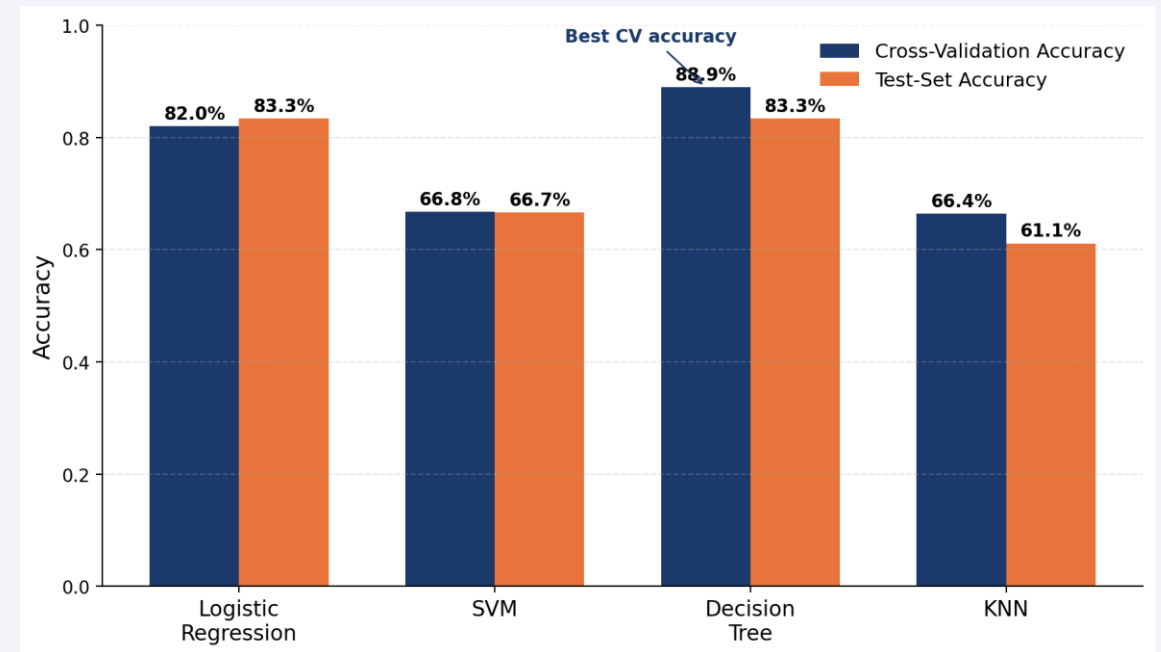


Section 5

Predictive Analysis (Classification)

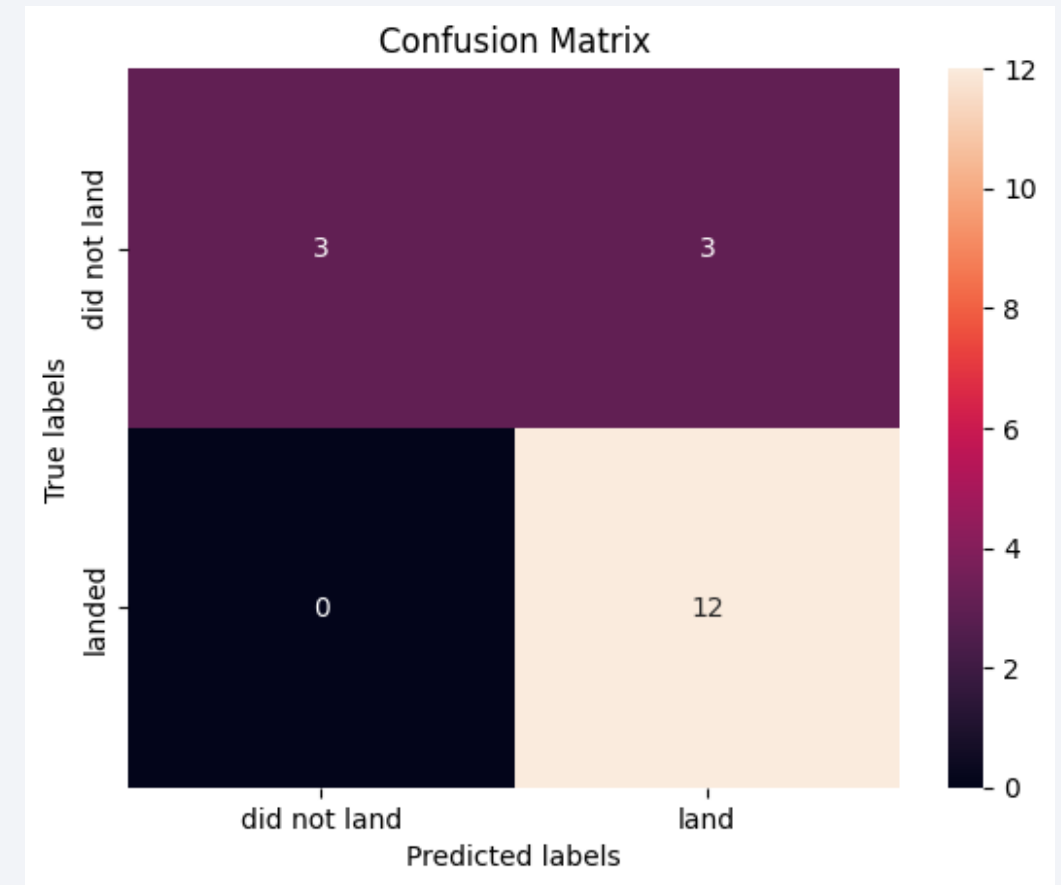
Classification Accuracy

- Logistic Regression: 82.0% CV accuracy, 83.3% test accuracy (C=1, penalty=l2, solver=lbfgs)
- SVM: 66.8% CV accuracy, 66.7% test accuracy (kernel=rbf, C=0.001, gamma=0.001)
- Decision Tree: 88.9% CV accuracy, 83.3% test accuracy — highest cross-validation accuracy of all four models
- KNN: 66.4% CV accuracy, 61.1% test accuracy (n_neighbors=3, p=1)
- **Best model: Decision Tree**



Confusion Matrix

- Confusion matrix shown is for the Decision Tree — the best-performing model (88.9% CV accuracy)
- On the 18-row test set: correctly identifies successful landings the large majority of the time
- Remaining errors are concentrated in false positives (predicting a successful landing when the true outcome was a failure) — the same pattern seen with Logistic Regression, suggesting this is a genuinely hard boundary in the feature space rather than a model-specific weakness



Conclusions

- Falcon 9 first-stage landing success rate improved steadily from near 0% (2013) to over 80% (2020), reflecting SpaceX's iterative engineering process
- Launch site, orbit type and payload mass are all meaningful predictors of landing outcome — KSC LC-39A shows the highest site-level success rate, and orbits like ES-L1/GEO/HEO/SSO/VLEO show the highest orbit-level success rates
- Flight number correlates strongly with success in LEO missions but not in GTO missions, showing that the "learning curve" effect is orbit-dependent, not universal
- A tuned Decision Tree classifier best predicts landing outcome from pre-launch features (88.9% cross-validation accuracy), narrowly ahead of Logistic Regression and clearly ahead of SVM and KNN
- Remaining prediction errors are concentrated in false positives — useful for a competitor to know: the model is more likely to over-estimate SpaceX's landing success than to miss a real success

Appendix

- Notebooks used in this analysis: Data Wrangling, Web Scraping (Wikipedia), EDA with Data Visualization, EDA with SQL, Interactive Map with Folium, Machine Learning Prediction — add your GitHub URLs here for peer review
- Not included in the source files for this deck: the SpaceX-API data collection notebook and the Plotly Dash application — add these before final submission
- Key datasets: dataset_part_1.csv (raw combined), dataset_part_2.csv (labeled, 90 rows), dataset_part_3.csv (one-hot encoded features, 80 columns)
- Full SQL query text for all 10 EDA-with-SQL tasks is available in the jupyter-labs-eda-sql-coursera notebook
- Best model hyperparameters — Decision Tree: criterion=entropy, max_depth=18, max_features=sqrt, min_samples_leaf=4, min_samples_split=10, splitter=random

Thank you!

